TASK 4: PTO AND HYDRAULIC ANCILLARY FAULT DIAGNOSIS

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 7 | v1.1

TIME	EVIDENCE	PROGRAMME POINT
90 minutes (adapt to workplace conditions)	**Practical | Knowledge | Documentation**	**Weeks 91-106 | Months 24-28**

Your task

Diagnose a PTO-driven hydraulic ancillary concern, such as a tail-lift or tipping system fault. Control stored energy, test the system and produce a safe repair and verification plan.

What you must do

1. Confirm operating complaint, equipment limits and authorised test conditions.
2. Isolate/support the ancillary and release stored pressure by the approved method.
3. Inspect PTO engagement, fluid, hoses, filters, pumps, valves and actuators.
4. Measure pressure/flow or electrical control signals at specified points.
5. Identify root cause and document repair, contamination control and proof testing.

What good evidence looks like

SAFE CONTROL

Mechanically support raised equipment and establish an exclusion zone; never rely on hydraulic pressure to support a load.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle ancillary or hydraulic rig
- Hydraulic pressure/flow and electrical test equipment
- Rated supports and exclusion controls
- Hydraulic diagram, specifications and PPE

SOW ALIGNMENT

Block 7 - Transmission, Driveline, Hydraulics, Ancillaries and Complex Repair

KSB evidence focus: K8, K10, B2; reinforced K1-K4, K6, K11, S1-S6, S8-S10, B3

Task reference: HV_L3_Block_7_Task_4_PTO_and_Hydraulic_Ancillary_Fault_Diagnosis

TASK 4: PTO AND HYDRAULIC ANCILLARY FAULT DIAGNOSIS

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Fluid level / condition		PTO engagement result	
System pressure		Pressure specification	
Flow / actuator time		Flow/time specification	
Control signal / valve result		Diagnosis / repair plan	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why can a supported load still be hazardous after the engine is stopped?

2. How do pressure and flow results help separate pump and restr

Knowledge check - continued

3. Explain contamination controls when opening a hydraulic system.

4. When must specialist or statutory inspection be involved?

5. What proves the ancillary is safe after repair?

Assessor/Workplace Mentor observation

- ☐ Controls suspended loads and stored hydraulic energy.
- ☐ Tests pressure/flow and controls safely.
- ☐ Plans functional/load proof and documentation.
- ☐ Uses hydraulic diagrams and specifications.
- ☐ Identifies root cause and contamination risk.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ☐ PASS
- ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 7 | Task 4

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. PTO will not engage

Checks: Check enable conditions, interlocks, switch request, control supply, solenoid/air pressure and mechanical drive.

Evidence: Record commands, pressure/voltage and safe engagement test with ancillary secured.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.